

LAMPHUN, Thailand

WMO: 483290

Lat: 18.5668N

Lon: 99.0384E

Elev: 979

StdP: 14.18

Time Zone: 7.00 (E07)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	54.4	57.0	46.4	48.2	77.4	49.2	53.7	75.2	7.6	77.8	6.6	78.2	0.0	60	0.289

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	23.4	101.9	74.3	99.7	74.0	97.5	73.8	80.0	89.9	79.5	89.1	79.0	88.4	3.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
78.0	150.8	83.0	77.3	147.5	82.3	76.9	145.2	81.9	44.6	90.2	44.0	89.5	43.4	88.7	88.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	8.0	6.7	50.0	104.8	3.2	2.2	47.7	106.4	45.9	107.7	44.1	108.9	41.8	110.5		
(5)			48.8	82.7	3.0	2.2	46.6	84.3	44.8	85.7	43.1	86.9	41.0	88.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	80.7	73.1	77.0	82.6	87.3	85.4	84.2	83.3	82.5	82.0	80.5	77.1	73.0
	DBStd	5.45	3.41	3.43	3.31	3.54	3.90	2.38	2.32	2.06	1.95	2.56	3.55	4.32
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	5	2	0	0	0	0	0	0	0	0	0	0	3
	CDD50	11198	716	756	1012	1118	1097	1027	1032	1007	960	946	814	713
	CDD65	5729	253	336	547	668	632	577	567	542	510	481	364	252
	CDH74	60876	2214	3662	6744	8864	7479	6562	6149	5459	4819	4141	2862	1921
	CDH80	27391	802	1818	3893	5250	3739	2773	2397	1946	1679	1463	1031	600
Wind	WSAvg	1.0	0.5	0.9	1.3	1.6	1.5	1.8	1.6	1.2	0.7	0.5	0.4	0.5
Precipitation	PrecAvg	43.0	0.4	0.3	0.9	2.0	6.6	4.7	5.5	7.8	8.5	4.3	1.6	0.4
	PrecMax	64.6	2.8	1.7	5.1	7.9	13.0	9.6	9.5	17.7	16.6	8.2	8.8	3.3
	PrecMin	28.7	0.0	0.0	0.0	0.3	0.5	0.9	2.7	3.3	4.1	1.0	0.0	0.0
	PrecStd	8.3	0.6	0.5	1.1	1.7	3.1	2.1	2.0	3.1	2.7	1.7	1.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.7	97.5	102.4	105.3	103.3	97.9	95.4	93.7	93.3	92.6	92.2	90.3
		MCWB	69.9	69.4	71.8	74.8	76.3	78.1	77.7	77.8	78.6	78.1	75.4	73.8
	2%	DB	89.1	94.9	99.9	103.1	99.8	94.7	93.1	91.8	91.3	90.8	89.9	87.9
		MCWB	69.5	69.3	71.2	74.5	76.7	77.9	77.6	77.7	78.4	77.6	75.3	72.5
	5%	DB	87.1	92.7	97.9	100.9	96.9	92.6	91.4	90.1	89.7	89.2	88.1	85.9
		MCWB	69.2	69.1	71.0	74.5	76.9	77.8	77.4	77.5	78.0	77.1	74.6	71.6
	10%	DB	85.0	90.3	95.7	98.6	94.1	90.7	89.6	88.5	88.1	87.4	86.0	83.6
		MCWB	68.7	68.9	70.8	74.4	77.1	77.5	77.1	77.3	77.8	76.7	73.8	70.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.3	76.3	79.3	79.9	80.5	80.5	80.2	80.1	80.5	79.9	78.3	77.2
		MCDB	85.7	91.0	95.1	97.4	92.7	91.0	89.5	88.4	88.8	88.6	87.4	86.3
	2%	WB	73.3	74.0	75.9	78.2	79.4	79.7	79.2	79.2	79.6	78.9	77.2	74.8
		MCDB	83.5	87.8	91.4	94.9	91.4	89.6	88.1	87.4	87.6	87.4	85.8	83.8
	5%	WB	71.8	72.4	74.5	77.2	78.7	79.1	78.6	78.6	79.0	78.2	76.2	73.3
		MCDB	81.6	85.6	89.6	93.1	90.7	88.4	87.2	86.5	86.7	86.0	84.2	82.0
	10%	WB	70.4	70.8	73.3	76.4	78.1	78.4	78.0	78.0	78.3	77.4	75.3	71.9
		MCDB	80.6	84.5	88.3	91.5	89.0	87.1	85.9	85.2	85.3	84.6	82.7	80.4
Mean Daily Temperature Range	5% DB	MDBR	26.7	29.9	28.3	23.4	16.6	13.5	12.8	12.6	13.1	14.6	19.0	22.7
		MCDBR	29.5	32.6	31.6	26.6	20.9	16.2	15.5	15.1	15.4	17.0	21.0	24.2
		MCWBR	11.3	10.6	8.7	6.0	4.6	4.2	4.1	4.1	4.5	5.3	7.3	9.4
	5% WB	MCDBR	23.5	27.2	25.8	22.5	17.0	14.3	13.6	13.4	14.0	15.2	17.6	20.6
		MCWBR	10.1	10.7	8.8	6.2	4.8	4.3	4.3	4.3	4.7	5.2	6.7	8.5
Clear-Sky Solar Irradiance	taub		0.436	0.513	0.671	0.667	0.504	0.439	0.427	0.442	0.475	0.517	0.443	0.437
	taud		2.199	1.995	1.637	1.708	2.172	2.386	2.418	2.359	2.248	2.085	2.258	2.223
	Ebn at Noon		266	251	217	220	256	270	274	272	263	247	262	261
	Edh at Noon		43	56	83	78	48	39	37	40	44	50	41	41
All-Sky Solar Radiation	RadAvg		1459	1702	1794	1910	1704	1480	1378	1363	1440	1411	1455	1371
	RadStd		74	60	116	112	144	106	105	86	62	80	102	81

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	+0.34	+0.41	N/A	+2	N/A	N/A
(47) Regional (17 neighbors)	+0.49	N/A	N/A	N/A	N/A	+0.27	N/A	N/A	+248	+250

Nomenclature: See separate page